

PERCEPTION OF SOUND VIA AUDITORY IMAGE ANALYSIS, MADE IN THE CONCEPTUAL CONTEXT OF THE PHILOSOPHY OF MUSIC

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Introduction

Since ancient times, it has been emphasized that while artists were musicians without knowledge of exact sciences, physicists and mathematicians were the people of science without knowledge of music. Analyses made by the latter placed all musical factors on the same level, in contrast to musician artists, who ranked musical elements in a completely different way than the scientists. For this reason, the interest in sound on the part of researchers from various scientific and artistic disciplines has been noticeable for ages. The question of how sounds created with instruments and the human voice (which itself is a building element of music) are perceived and interpreted by hearers has been the subject of deliberations for many centuries. What is sound – what is its genesis, construction, and structure? This article focuses on the perception of sound via analysis of an auditory image in the context of the philosophy of music.

Problematic aspects of a definition of sound

Two of the most frequently encountered definitions of sound are: *an acoustic wave that propagates through an elastic medium*, or *an auditory experience caused by such a wave* [1, p. 28]. A person can differentiate between various qualities of sound, among others pitch, duration, volume, and timbre. *Composing may be based on manipulation of these elements perceived by our ears. (...) In the past, it was unthinkable that what we hear as one quality, is in fact a quantity. Nowadays, we know that almost each sound perceived by us consists of numerous overlapping acoustic waves* [2, p. 45]. A complex sound is constituted by many harmonic and non-harmonic tones; thus, the significance of timbre is the most difficult to capture because many other circumstances, such as transients, formant, and other elements influence the timbre. While frequency of the lowest component tone determines a sound's pitch, distribution and intensity of the remaining waves determine a sound's timbre. Owing to this phenomenon, we can differentiate the timbre of, for instance, a violin, grand piano, harp, clarinet, trumpet, oboe, bassoon, or symphonic orchestra [3, pp. 940-942].

Perception of sound in terms of auditory image analysis

Perception of a sound is related to the functioning of two types of processes in the human auditory system: the sensory processes and the cognitive processes. The sensory processes are a set of mechanisms responsible for processing acoustic waves entering the ears into nerve impulses, coding data in the form of impulses

regarding physical characteristics of the received sounds, and transferring the nerve impulses to the hearing center in the cerebral cortex. The functioning of the sensory processes connected with hearing depends on the physiological condition of the human auditory system. It is usually assumed that no significant differences exist in these mechanisms between persons having physiologically normal hearing. This means that all people whose auditory system is in good condition, regardless of pursued profession or musical education, receive incoming sound stimuli in a highly similar manner, using the same mechanisms.

The cognitive processes are related to information coming from hearing and other senses that are processed by the mind. This processing takes place in the central nervous system and consists of receiving information from the surrounding environment, its storage and modification via reflection, and its re-introduction to the surroundings in the form of a reaction (i.e., behavior). A person uses the cognitive structures pertaining to sound that are received in this manner to generate a mental image through the modification and analysis of sound stimuli. The cognitive processes include, among others, attention, perception, memory, and thinking [4, p. 178, 278, 320, 420].

The cognitive processes also depend on a variety of factors connected with the past experiences of the person receiving sensory stimuli. In the case of auditory stimuli reception, the cognitive processes are significantly affected by experience gained during musical education and professional experience acquired in the musician's profession. For this reason, persons with a similar physiological condition of the hearing organ who have had various experiences connected with sound perception may differ in their sensitivity to changes of sound or modifications of its timbre, and may receive auditory images of different properties and structure when the hearing organ is stimulated with the same acoustic signal.

Hearing is based on the recognition of particular characteristics of sound of an auditory stimulus. In traditional research on the principles of sound perception, researchers use specialized methods of hearing assessment, allowing them to obtain a description of dependencies between physical characteristics of sound and characteristics of a sound sensation received by the listener [5, pp. 27-36].

An auditory image is a term referring to the receipt of sound stimuli, namely, sound perception. The analysis of this image's features is understood as focusing the recipient's attention on certain physical characteristics of a sound wave which may bear certain similarities. Perception streams are created in the listener's mind as a result of the analysis of an auditory image pertaining to sequences of sounds. The ongoing sound processing causes integration of sounds into one stream or segregation of sounds into separate streams (formation of one or many melodic lines). The integration and merging into a common structure also occur in cases when sounds are heard simultaneously (creation of a harmony) [6, p. 861].

Auditory image analysis is a process that includes the scrutiny of sound stimuli coming from the listener's environment and development of mental

representations created as a result of stimulus influence on the recipient's mind. A highly significant element of the creation process of sound representations is the determination of which components or elements of a sound stimulus deliver the most significant information that allow a mental figure to be created (for instance, the main pattern of melody or harmony) in the recipient's mind. The process of auditory image analysis is targeted at identifying independent descriptions of each of the objects the recipient distinguishes through the analysis of sounds from the environment [7, p. 20].

Exploration of auditory image analysis shows that the term applies only to the phenomenon of perceiving (in themselves), while the processes creating auditory representations are understood as an independent system that organizes elements and items into a whole (for instance, complete creation of a melodic line) [8, pp. 28-29]. The basic field of experience shows the received stimuli at the stage of entirety, which results from our previous practice. Multiply repeated experiences cause the development of similar mechanisms for stimuli processing in the case of receipt of coherent and similar content with related elements [9, pp. 145-146]. It is worth noting that this approach is related to a conviction about the existence of certain organizing mechanisms that create a larger – semantic – role in perceptive experiences. Like the visual system, the auditory system also analyzes other variants and possibilities of solving a particular auditory puzzle in a different manner, but still preserves one version as the active one at a given moment [7, p. 65].

A reaction via the ear, which is a passive organ in its nature, in contrast to the sharp and selectively active eye, in some way departs from the present epoch of advanced industry and its cultural anthropology. For this reason, the acoustic perception preserves comparably more qualities of the past pre-individual communities than the visual perception [10, p. 103].

Our visual perception seems to be slightly simpler than our auditory perception because sight is superior to hearing. If we analyze a painting, we have unlimited time to perceive it. We can analyze each point and dot from a different angle and in different light, each time noticing a different element which we had not noticed before. But in the case of live music or music replayed from an electro-acoustic set, our scope is limited because melody is a sound structure constrained within time. It cannot be perceived at a pace other than that of the played or replayed music; therefore, music imposes the pace of perception. This time frame does not exist in the visual domain. A sound cannot be slowed down, because this alters it, and doing so could modify one's interpretation of a melody. Another significant difference between sight and sound is that a painting or a sculpture can be partially perceived "all over again." Unfortunately, in the case of replaying music, a listener must rewind the track to the beginning of a given phrase to be able to understand the sequence of sounds occurring in it. Therefore, musical perception poses more problems and thus is more difficult to understand.

Constructing a separate description of each of the perceived sources of sound consists of analyzing the input signal using the heuristic process. Sound

information created in this way, that is based on the structure of a sound influencing the listener, results from the fact that actually a set of sounds (that is, the total of basic sounds) reaches the human ear [11, p. 34]. The analysis of an auditory image reveals how the auditory system sorts out and constructs an auditory scene based on complex data in the form of overlapping stimuli [3, pp. 940-942]. By analyzing the data, the brain may qualify these stimuli to a given sound stream (one melody) in the integration process or treat them separately as assigned to various streams (several various melodies – for instance, polyphonic music) in the segregation process, thus deciding whether the stimuli belong to one sound stream or not [12, pp. 13-14].

Considerations pertaining to the essence of sound perception

By listening to music, we receive its inner spirit, which influences us. By listening to its orders and admonitions, the entire world of sensations, passions, and experiences changes with music; we are thereupon permeated with the beauty of sensations and feelings. According to Adorno, there are two different types of music perception: the unnatural ones, co-created by history, which can be assigned to two main sets of a character's features. The expressive-dynamic type derives from singing: that is, vocal music. This type is *oriented towards overcoming time by its filling and in its highest expressions processes extra-musical lapse of time into dynamism of a musical process* [13, p. 250], while the rhythmical-special type results from beating a drum. This second type includes division of time into two equal sections which simultaneously liquidate and spatialize time.

Both types differ significantly from each other due to social alienation, which causes the separation of a subject from the object. *Musically, everything subjective is presently threatened with contingency, and everything objective – with alienation and tyrannical severity of bare existence* [13, p. 250]. The idea of great music was mutual penetration of both manners of receipt, which referred to the subjective (humanized) and the objective (measurable) manner of sound reception, which determined appropriate proportion for a given composer, responsible for a great piece of art.

I want to find the truth which is the truth for me. And for what could I use the so-called objective truth, if it would not have deep meaning for me and my life? [14, p. 88].

The philosophy and the theory of music are still trying to classify what is a consonance or a dissonance. Let us not forget that the images of consonance, dissonance and sound perception have been changing throughout the centuries. *What do I hear?... What exactly do you hear?* [15, p. 107]. These questions refer not only to the external source of a sound, but also to the hearer's perception of what that sound signifies on a subjective or psychological level.

Objectivity consists in making use of physics, where the laws of

the observable world reign supreme, and the effects of music are only skin-deep. Subjectivity, on the other hand, ignores measures and experiences used in describing a physical object [15, p. 107]. Subjectivity ensures an approach towards the subject, to its sphere of experience (mental activities) through the interaction of sounds, and thus through musical perception. Each person is different; hence, is the subjective approach appropriate? Each person may interpret the same acoustic stimulus in a different or even similar, but never identical manner. Perhaps objectivity is a reasonable intention, and the target which should be pursued after all? Nevertheless, music is composed for people, not machines, and there is no musical gift that influences the perception and sound reception without increasing human sensitivity, despite our unique and differing responses. Music must draw from subjective feelings; otherwise, it would become a mechanical and insignificant blend of sounds.

To thoroughly examine various types of acoustic phenomena, subjective research methods are replaced with entirely objective methods, where a given issue is examined with the use of appropriate devices. So, can digital devices be humanized? Is the interpretation of their data reliable and consistent with humanity? Are mathematical data compatible with the subjective feelings of each living entity?

Acoustic criteria cover subjective as well as objective aspects. The subjective aspects refer to acoustic space and sharpness of timbre that people feel or sense but that are not directly or empirically measurable. In turn, the objective aspects are subject to exact analyses (i.e., are measurable by appropriate instrumentation), which can be described using physical descriptions of sounds and parameters of the acoustics field [16, p. 244].

Perception of a sound is not a mathematical mechanism which can be categorized and assigned a meaning: good/bad, or appropriate/inappropriate. In the scope of assessment, one should use objective methods and criteria with accompanying subjective elements [16, p. 244]. Appropriate and, at the same time, satisfying acoustic sensations are a collection of subjective sensations and objective (measurable) parameters of the acoustics field.

In *The Republic*, Plato assigns (among others) pedagogical meaning to music and to the ability to listen, and compares particular states of the soul to specific musical scales. In his treatise on the theory of music (*On the sensations of tone as a physiological basis for the theory of music*), H. L. F. Helmholtz claims that musical beauty has its origin in conformity with the elemental laws of nature. Riemann presents his views in numerous monographs (*Die Elemente der musikalischen, Ästhetik*, and *Katechismus der Musikästhetik*) in which he emphasizes the importance of sound performances as the basic subject of research, discusses functions of sound imagination, and delineates the role of intellectual activity in an aesthetic experience [17, pp. 6-7]. In contrast to these viewpoints, for the first time in the history of aesthetic thought, in *Critique of Judgment* Kant puts forward a thesis that music cannot contain or express conceptual content.

The presented philosophical theories and acoustics research clearly indicate that there is no such thing as “loneliness of sound.” Hearing is not sufficient; listening must also be made comprehensible. Then, by opening our souls, we can fully experience a musical piece of art. Intellectual activity – that is, an understanding of music – is a significant matter that allows for appropriate musical perception. We should also pay attention to experiments conducted by many researchers which indicate that the members of a musically educated society perceive sounds in an entirely different manner compared to persons without musical education [18, p. 605; 19, p. 1, 9; 20, p. 9; 21, p. 659; 22, p. 14106; 23, p. 199; 24, p. 167].

Music sets and keeps strong principles. These principles apply not only to listeners, but to composers as well. In fact, the ear of an educated artist will “trace” and examine as much matter as is within its reach. Conscious listening to sound structures may become the reason and inspiration – with simultaneous occurrence of an emotional (and/or spiritual) element – for the creation of musical works [25, p. 89]. *When an artist enters a state of spirituality, he or she is inspired and able to create. Not only works, but also new sensations and strong emotions are born out of inspiration. As one steps outside the world of phenomena, a need to create new connections is born within, an “unsatisfied longing for transcendence” and contemplation of Higher Beauty* [26, p. 157]. When Beethoven wrote his last works, including the Ninth Symphony, he had completely lost his hearing. One might conjecture that in his later years, isolated from social discourse by his deafness, Beethoven “created new connections” and achieved inspiration and a “contemplation of Higher Beauty” through his musical creations.

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